W10

Species distribution models

STATISTICS GIS & MAPPING

A species distribution model (SDM) relates where a species has been seen to environmental conditions in order to predict where habitat is suitable. Today you build an SDM for woodlark with the cleaned Week 8 records, compare it with MaxEnt and evaluate it honestly.

DURATION	About 2 hours.
PACKAGES	`terra`, `sf`, `maxnet`, `tidyverse`
FILES	`week10_species_distribution_models.R`, `week10_solutions.R`, `gbif_lullula_arborea_clean.csv`, `dem.tif`, `landcover.tif`, `tavg.tif`, `prec.tif`, `montfred.gpkg`
DATA	205 thinned presences and 3,000 background points. tavg.tif and prec.tif imitate WorldClim layers at a local scale.

Learning outcomes

- prepare predictors, including landscape ones (proportion of habitat within a radius);
- detect collinear predictors and choose among them;
- fit an SDM with a GLM and with MaxEnt (maxnet) from presences and background;
- evaluate with AUC and spatial-block cross-validation;
- turn a suitability map into a presence/absence map with a threshold.

1 · Predictors and collinearity

Birds respond to the surrounding landscape, not to a single 30 m cell. So we calculate the proportion of heathland and meadow within about 250 m with `focal()`. Before modelling we look at the correlations:

	elevation	tavg	prec	open_250
elevation	1	-0.99	0.99	0.75
tavg	-0.99	1	-0.98	-0.74
prec	0.99	-0.98	1	0.75
open_250	0.75	-0.74	0.75	1

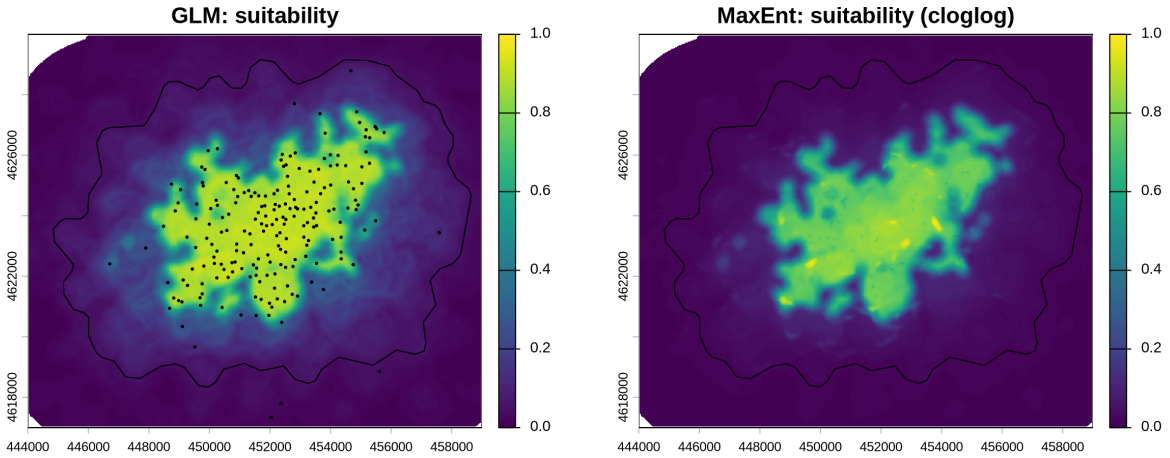
Temperature and precipitation are almost a transformation of elevation ($|r| > 0.9$): using all three gives unstable coefficients and no new information. We keep elevation, slope and proportion of open habitat. The same often happens with WorldClim; a common rule is not to combine predictors with $|r| > 0.7$.

2 · Presences and background

GBIF only gives presences. **Background points** are not absences: they describe the environment available in the study area (the park plus 3 km). The model compares where the species is with what is available. In the GLM we give presences and background the same total weight.

```
w <- ifelse(dat$pa == 1, 1, sum(dat$pa == 1) / sum(dat$pa == 0))
m_glm <- glm(pa ~ elevation + I(elevation^2) + slope + open_250,
             family = binomial, weights = w, data = dat)
suit_glm <- predict(env, m_glm, type = "response")
m_mx <- maxnet(dat$pa, dat[, c("elevation", "slope", "open_250")])
suit_mx <- predict(env, m_mx, type = "cloglog", na.rm = TRUE)
```

Proportion of open habitat is the strongest predictor ($p < 0.001$); elevation adds little once it is included. That makes sense: woodlark is a bird of open ground with scrub, and here open ground is on the summits.



Habitat suitability from the GLM (left, with presences) and from MaxEnt (right).

3 · Honest evaluation

AUC is the probability that a random presence scores higher than a random background point (0.5 = random, 1 = perfect). If test data are picked at random, they sit next to training data and AUC is inflated by spatial autocorrelation (Week 9). With 3 km **spatial blocks**, each whole block goes to training or testing, as when predicting to a new place.

Cross-validation (5 folds)	Mean AUC
Random	0.91
3 km spatial blocks	0.89

With the threshold that maximises sensitivity + specificity (0.55), about 3,080 ha of the park (28%) is suitable habitat.

Limits of an SDM

A presence–background SDM predicts *relative suitability*, not probability of presence or abundance. It inherits the biases in the records (Week 8) and knows nothing about dispersal, competition or history. Predicting to other regions or to future climates outside the data range is extrapolation.

Exercises

Answers in `week10_solutions.R`.

1. Fit the GLM without the quadratic term and compare AIC and the response curve.
2. Replace elevation with temperature. Does the map change? Why?
3. Correlation between the GLM and MaxEnt maps.
4. Block CV with 1.5 and 5 km blocks: how does AUC change?
5. What percentage of the park is suitable?

Before Week 11

- Install `install.packages("adehabitatHR")`.
- Optional reading: Elith and Leathwick (2009), *Species distribution models*, Annual Review of Ecology, Evolution, and Systematics 40: 677–697.